What a median alert lead time measures: a reporting gap in continuously evaluated clinical models

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Abstract

A median alert lead time is readily read as a statement about the model: how much warning it gives. Yet it is also shaped by the cohort, because a first threshold crossing can only be found within each patient's available pre-onset window. A single median cannot separate model behavior from the distribution of those windows, making this a reporting problem before it is a modelling one. That a lead time cannot exceed its search window is definitional; what has not been measured is how much of a reported number that window accounts for. Across three perioperative cohorts and five model families, at thresholds transported unchanged from a development validation split, median lead rises from the shortest to the longest realized pre-onset window group in all fifteen model-by-cohort combinations, by 26 to 54 minutes. Read beside its operating point, the longest median lead here, 24 minutes, comes from the combination with the lowest case-level specificity, 0.406. Two natural repairs leave the dependence in place. This is an empirical measurement-and-reporting case study; we do not measure what clinicians understand from the term. A median lead remains a descriptive interval among detected cases, but on its own it does not support a model-intrinsic or cohort-comparable reading of warning capability. We recommend reporting the pre-onset window distribution, the operating point, and a window-conditional or explicitly qualified horizon-specific timing measure beside any single median, and set out four candidate disclosures, a task-specific complement to TRIPOD+AI.

1 Introduction

Why a warning interval matters here. Mean arterial pressure is the average arterial pressure over one heartbeat, and it is the number an anesthetist watches most closely during an operation. When it falls below roughly 65 mmHg and stays there the episode is intraoperative hypotension, and how much a patient accumulates is associated with kidney and myocardial injury after surgery [Walsh et al., 2013, Salmasi et al., 2017, Wesselink et al., 2018, Sessler et al., 2019]. The responses are ordinary anesthetic care, and none acts the instant it is chosen, so a warning interval is worth something to the extent that it is time in which a clinician can choose and deliver one. That is why monitors here are marketed and evaluated on how far ahead they alert [Hatib et al., 2018, Davies et al., 2020, Wijnberge et al., 2020, Maheshwari et al., 2020]. This paper is about that reported interval and not about physiology: the outcome is defined on mean arterial pressure, which is also a model input, and Section 4.2 says what that does and does not license.

A model that is read continuously is asked how early it warns, and the **median alert lead time** is the statistic reported in answer across acute care: the interval between the first minute a case's score crosses an alert threshold and the onset of the event. We have not surveyed how often it is reported, so what follows is a multi-cohort case study rather than a field-level audit. It appears across acute

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care, for sepsis [Wong et al., 2021], for acute kidney injury [Lee et al., 2026], and for intraoperative hypotension, where a commercial index is marketed on the interval between its alert and the event. The statement that a system has a **median alert lead time of 24 minutes** invites the reading that it will usually give around twenty-four minutes of warning, and that reading is what makes the number worth reporting. We study the correspondence between the statistic and that invited reading; we do not measure clinician comprehension directly.

The definition is the same in every one of those settings, and it carries a dependence that its name conceals. Score every eligible minute, compare against a threshold, and record the interval between the first crossing and the event. The first crossing is searched for inside the interval the patient was observed, so the lead is bounded above by the observation window. The available run-up to onset is set by event timing relative to eligibility, by monitoring practice and by case mix rather than by the model.

What is already known, and what is new here. Criticism of early-warning evaluation is not new: the C-statistic has been argued uninformative for early warning scores, with per-alert and workload measures proposed instead [Romero-Brufau et al., 2015]; monitor alerting has been quantified as a per-hour burden [Paine et al., 2016]; choice of evaluation sample moves reported performance for this exact task on these public data [Yang et al., 2025]; and cross-center evaluation separates learned methods from pressure-derived rules far less cleanly than single-center reports suggest [Chaari et al., 2026]. That a lead time cannot exceed the window it is searched in is likewise definitional, and Section 3.3 treats it as such. What is not established is **how much of a reported lead time is the window**, and the substitutes those critiques recommend are what Section 6.2's second item asks for.

We make four contributions. We measure the dependence, in the metric's own units and against two explicit references rather than against the ceiling (Section 3). We report the operating point beside the lead, which no lead-time statistic carries, and find the two run opposite to one another across combinations (Section 4). We implement the two repairs the first two sections invite and neither works, while a window-conditional statistic that does work disagrees with the median (Section 5). And we map the case study onto an author-completed TRIPOD+AI [Collins et al., 2024] checklist, which did not request the disclosures that would have made any of this visible, and propose four candidate additions for continuously monitored models (Section 6). The argument is about a reported quantity, not about how well any model discriminates, which is not in dispute here.

2 Setting

Task and outcome. Intraoperative hypotension is mean arterial pressure below 65 mmHg sustained for at least three consecutive minutes, the conventional threshold in this literature [Walsh et al., 2013, Salmasi et al., 2017, Sessler et al., 2019]. Mean arterial pressure is also one of the models' input channels, so the label is computed from a signal the models read; Sections 4.2 and 7 say what that does not license.

Cohorts. Development used MOVER's EPIC arm; we evaluate on its held-out test split (1,404 cases, 786 with a sustained event) and on two external cohorts, VitalDB [Lee et al., 2022, Lee and Jung, 2022] with 2,804 cases and 1,007 events, and the SIS-derived arm of MOVER [Samad et al., 2023a,b] with 2,641 cases and 1,409 events. MOVER (SIS) releases no usable subject key, so cases are the clustering unit there and subjects in the other two; all intervals in this paper are 95% cluster bootstrap on that unit unless a caption says otherwise.

Models. Five families, spanning the range a clinical team would realistically compare: a transformer over irregularly sampled triplets [Tipirneni and Reddy, 2022], referred to throughout as the ITE Transformer, an LSTM, XGBoost, random forest and logistic regression. Two read the window as a sequence and three a per-minute feature vector; Appendix A.1 has the architectures and training. No model is proposed for clinical use and none of what follows is a ranking.

Deployment simulation. Every model is read forward at **every eligible minute of every case**, using only measurements recorded strictly before the prediction minute and after a source-specific exclusion for the induction artifact (Appendix A.1), so a 1,404-case cohort yields tens of thousands of scored case-minutes from cached data after the fact.

Operating points. An **operating point** is the score threshold at which the system is read as alerting, with the sensitivity and specificity that follow. **Case-level sensitivity** is the share of event cases that

alert at some point in their alert window and **case-level specificity** the share of control cases that never do. Calibration and thresholds were fitted on the development cohort's validation split at an 80% case-level sensitivity target and then **transported** unchanged, applied to every other cohort as fitted rather than re-estimated there. Nothing here reselects a threshold, split, seed or operating point, and no model is retrained or re-scored.

The alert rule and the window it searches. A case alerts if its score reaches the threshold at any minute in the 60 minutes before the event, and its lead time is the interval from the first such minute to the event. The interval the rule searches is the case's **available pre-onset evaluation window**: its realized observable run-up to onset, after the induction exclusion, capped at the 60-minute alert window. A short window is therefore not a briefly or incompletely monitored case. For an event case it means the event arrived early in the record, so the run-up *is* the time to onset and there is no missing data to recover. An alert already active at a case's first eligible minute is **prevalent**; one that begins later is **incident**. Cases without an event take a sampled pseudo-onset drawn from the event cases' timing distribution, so controls are scored over a comparable window. Every lead time reported here uses this rule unchanged.

Ethics. An institutional review board granted an exemption from review and waived consent for this secondary analysis (Appendix A.1).

3 Lead time tracks the available pre-onset window

3.1 The number moves with the window, in minutes

This section splits detected cases by the length of their available pre-onset window and reads the median lead separately in each group, which puts the dependence in the metric's own units.

The reported lead time moves with the group (Table A9), the same way in every cell. Median lead rises from the shortest to the longest realized band in **15 of 15** model-by-cohort cells, by **+26 to +54 minutes**, every interval excluding zero (Table A1), an endpoint contrast of **0.51 to 1.02 minutes of lead per minute of window difference**. A single headline figure therefore describes the cohort's observation-length distribution as much as the model: across its three cohorts the transformer reports 5 to 9 minutes of median lead for its shortest-observed band and 46 to 59 for its longest, and the pooled 17 to 24 minutes sits between them because that is where the median case happens to fall.

Detection is itself a function of window length, so we report that rate rather than conditioning on it silently. Re-cut on all event cases, the detected share moves with window length in every cell, by 0.105 to 0.556, falling in the two families that read the window as a sequence and rising in the three that read a per-minute vector (Table A5). For the transformer the selection runs *against* the reported gradient, which survives it; for the tabular families it runs with the gradient, so part of theirs may be survivorship.

3.2 The same fact as a ratio, in every family

Section 3.1 gives the dependence in minutes; as a fraction of each case's own window it separates what is universal here from what is model-dependent.

The gradient is universal; its magnitude is not. The median ratio of a case's lead to its own observed window is 0.455 to 0.944 and the Spearman correlation between lead and window length is 0.501 to 0.906 across all fifteen cells, again with no interval covering zero. The transformer is the limiting case at 0.899 to 0.944 with $\rho = 0.706$ to 0.906, its first alert falling 90 to 94% of the way back to the start of the window it was given; the mildest cells are logistic regression in VitalDB (0.455) and XGBoost in VitalDB (0.467), where the first alert falls closer to the middle of the window than to its start. The direction holds in every family and cohort while the strength ranges from near-total to moderate, so a claim that the quantity *is* the window fits the transformer and overstates the tabular families.

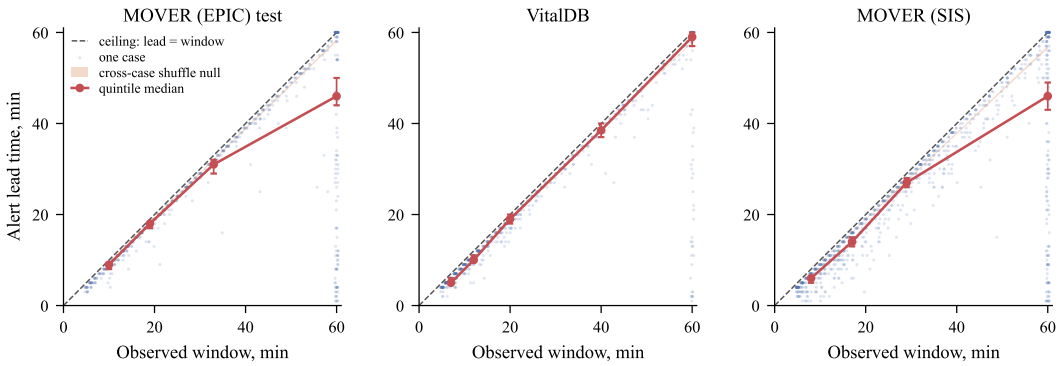


Figure 1: **Alert lead time against the available pre-onset window.** Every detected event case under the ITE Transformer, three held-out cohorts. The dashed diagonal is the ceiling: no crossing can be found outside the case's own window. The shaded band is the 2.5th to 97.5th percentile envelope of the cross-case shuffle reference, not a per-case prediction interval. Red markers are the band medians of Table A9 with 95% cluster bootstrap intervals; all five families are Figure A1.

3.3 Explicit references for the ceiling dependence

Part of the relationship is built into the definition: a lead time cannot exceed the window it is searched in, so the ratio is bounded in [0, 1] by construction and positive dependence can arise without patient-specific timing. A bound does not specify a distribution, so we use two explicit references rather than treating the diagonal as a null.

The interpretive rule comes before the numbers, because it decides what they can mean. Both references are biased *toward* agreement with the observed values, in the direction that favors our own reading, so each is an upper bound on what a window-only process could produce. A reference that reaches or exceeds the observed value is therefore the expected and uninformative outcome; what would be informative is one falling clearly short, and neither does.

A continuous uniform reference has median 0.5. Placing a pseudo-crossing uniformly on each case's own window gives a reference median of 0.518 to 0.534, which the observed ratio exceeds by **+0.372 to +0.421** for the transformer. It does not exceed it everywhere: in **2 of 15** cells the observed ratio sits *below* the reference and in one more it is indistinguishable from it, all three VitalDB cells (Table A2).

A cross-case shuffle preserves the aggregate relationship (Figure 1). Each recipient keeps its window and reference time but receives another case's whole score sequence, aligned by elapsed window time, so generic elapsed-time shape survives while the match between trajectory and recipient is destroyed. Permutations are derangements, 1,000 per cell, so no case keeps its own sequence. The shuffled median ratio is **0.72 to 0.97** and Spearman **0.63 to 0.94**, so the reference exceeds the observed ratio in **15 of 15** cells, which under the rule above is what an upward-biased reference is expected to do. Under a length-matched variant the observed correlation is **not distinguishable from the reference in 9 of 15 cells**, and the reference is the larger in 6 of 15 (Table A2).

Under these deliberately conservative references the observed association did not exceed what a window-only reassignment can produce. That is not the same as showing that a window-only process generates it, nor that patient-specific information is absent. Reassignment also changes which event cases are detected, and both variants bias upward, by selecting longer, already-settled donors or holding a short donor's last value. They bound the dependence rather than estimating a case-independent process.

4 Score behavior behind the reported lead

Section 4.1 identifies a direct mechanism for one model. A cross-family property of the score paths is consistent with it without establishing its cause, and is reported in Appendix A.4 beside Table A3.

4.1 The transformer is above its own alert threshold almost immediately, in event and control alike

This section asks what the score is doing at the minutes the alert rule reads, and what the operating point behind the reported lead looks like.

Within one minute of its first scored minute the transformer's median score is already above the threshold its operating point uses, and it is there for event and control cases equally (Figure A5).

In MOVER (EPIC) the median score rises from 0.530 at the first scored minute to 0.938 one minute later in control cases, and from 0.541 to 0.951 in event cases, against a transported threshold of **0.894**. By minute five the two groups are at 0.973 and 0.975, a separation of 0.002 in a score whose scale runs to 1. VitalDB behaves the same way, reaching 0.966 and 0.973. MOVER (SIS) settles more slowly, at 0.833 and 0.906 by minute two and 0.942 and 0.962 by minute five, but it too clears the threshold and it too does so in both groups.

The consequence for the metric is direct. A case whose score settles above the threshold within its first monitored minutes will register its first crossing as soon as its alert window opens, whatever else happens, so its lead time is whatever that window holds. That is the near one-for-one contrast in Table A9.

No other family does this (Figure A5). Sustained elevation is visible in the share of monitored minutes at or above threshold: **20.0 to 52.3% of the transformer's event minutes and 2.5 to 18.3% of its control minutes, against 0 to 26.7% and 0 to 1.7% for the other four families**.

The obvious objection is that this model simply alerts continuously, and reporting the operating point beside the lead shows that it largely does (Table 1). Its case-level specificity is 0.406, 0.379 and 0.467, so it alerts on 53 to 62% of control cases, and its positive predictive value of 0.625, 0.380 and 0.585 sits 0.065, 0.021 and 0.051 above the prevalence it is measured against. This is not a competing explanation of Section 3 but the same fact: a score above threshold most of the time alerts at the first opportunity, which is what makes the lead the window. The point is that **the reported lead carries none of it.** A median lead of 24, 17 and 22 minutes is the largest of the fifteen combinations and comes from the one with the least usable operating point.

That holds beyond one model. **Across the fifteen combinations the reported median lead runs opposite to case-level specificity** (Spearman -0.803, permutation $P = 0.0008$, $q = 0.006$; 86 of 99 untied pairs), and opposite to PPV in excess of prevalence at -0.719. Within a cohort the ordering is 9 of 9 untied pairs in both MOVER cohorts and 8 of 10 in VitalDB, but five combinations per cohort is thin and none survives adjustment alone, so the pooled figure carries the claim and cohort confounds it. Two further limits belong with it. The unit is a combination and not a patient, so these are permutations of the pairing rather than cluster bootstraps. And a transported threshold does not deliver the same realized sensitivity everywhere: it runs from 0.461 to 0.849 across these combinations (Table A4), so this is a descriptive association among the operating points as transported, not a sensitivity-matched ranking of models. What it supports is narrower and still worth stating: a reader ranking these systems by the number they publish would not be ranking them by how usable their operating points are.

Table 1 | Case-level operating point behind each reported lead time. ITE Transformer, three held-out cohorts, at the transported 80%-sensitivity operating point every lead time in this paper uses. `Min. above thr.` is the median share of an event case's monitored minutes at or above threshold. Event prevalence is a property of the cohort, not the cell: 0.560, 0.359 and 0.534. **All five families, every interval, the number needed to evaluate, the control-side burden and PPV in excess of prevalence are Table A4**, where the cross-family claim below is checked.

Model	Cohort	Sens.	Spec.	PPV	Min. above thr.	Lead, min
Transformer	EPIC test	0.777	0.406	0.625	0.523	24
Transformer	VitalDB	0.679	0.379	0.380	0.444	17
Transformer	MOVER SIS	0.657	0.467	0.585	0.200	22

4.2 What these findings do and do not license

Both findings are properties of a pipeline as deployed with this scoring policy rather than claims about what any architecture must do; Section 7 states the scope. Neither is a claim about physiology. The outcome is a sustained excursion of mean arterial pressure and mean arterial pressure is one of the model's input channels, so the model has direct access to the quantity the label is computed from, and nothing here shows that a score anticipates a physiological process.

5 Two natural remedies, and why neither works

Two fixes follow naturally from Sections 3 and 4, and neither works. A metric with a plausible-looking repair that fails is a worse problem than one with none, because the repair gets adopted.

5.1 Prevalent alerts are not the explanation

The first thought is that the metric is dominated by cases already above threshold at their first eligible minute. Such cases are **2.6 to 25.5%** of detected event cases across the fifteen cells, and removing them to recompute the median lead on incident alerts alone costs **0 to 3 minutes in all fifteen cells**, the transformer included (Appendix A.11). The relationship of Section 3 is therefore carried by the ordinary cases. Prevalence stays this low alongside the settling of Section 4.1 because most detected event cases have an available window shorter than 60 minutes, so it opens while the score is still settling: prevalence and a warm-up crossing are the same behavior on either side of that minute.

5.2 Suppressing the first minutes of monitoring is not the fix

The second thought follows from Section 4.1: if the score is unreliable while it settles, discard alerts raised during that period. We implemented exactly that, discarding every minute at or before N minutes of elapsed monitoring and then applying the unchanged threshold.

At $N = 5$, across all five families and three cohorts, the rule costs **3.6 to 6.8 percentage points of case-level sensitivity** and returns **0 to 3.2 percentage points of specificity**. Median lead changes by **−3 to +1 minutes**; in two of the fifteen cells it *increases*, because discarding early minutes removes some short-window cases from the detected set entirely and shifts the median. The trade is poor in every cell, and it does not address Section 3: the surviving alerts still fire at the first opportunity within whatever window remains.

The comparison holds the transported threshold fixed, so it is deliberately unmatched on sensitivity; reselecting a threshold is the one thing this paper does nowhere. The direction is not expected to reverse under matching: suppression removes alerting minutes and cannot add crossings, so after lowering the threshold to restore sensitivity, any specificity gain cannot exceed the fixed-threshold gain above.

We therefore do not recommend the rule. The problem is not that early minutes are noisy. It is that the reported quantity is coupled to the length of the window it was measured in.

5.3 What partially helps, and what remains

Both repairs above change the alert rule and leave the estimand alone. The remaining option reverses that: condition on window length, or treat the first alert as a time-to-event whose follow-up ends when the window does, the landmarking treatment of a prediction made at a fixed point in a run-up [van Houwelingen, 2007]. Both are computable from the same cached scores, so we report them rather than only recommending them (Tables A6 and A7).

A horizon-specific detection rate reduces the dependence rather than removing it. It makes the horizon explicit and conditions the denominator on event cases observed for at least X minutes before onset. That removes one ambiguity of the pooled median, but window length is still not constant: cases observed longer than X can contribute different amounts of pre-horizon alerting opportunity. The rate runs 0.278 to 0.692 at a 10-minute horizon and 0.173 to 0.547 at 30 minutes, declining smoothly with the horizon, a shape a single median suppresses (Table A6). Restricted instead to cases observed for the whole 60-minute window, so the window is a constant, the transformer reports the longest median lead of the five families in all three cohorts, 47, 60 and 57 minutes,

while detecting only 0.441, 0.247 and 0.304 of them against 0.762 to 0.910 for XGBoost and the random forest (Table A7): conditioning turns its reported advantage into a deficit.

Both remain partial. The full-window restriction is the stricter window-constant estimand, but it changes the estimand and the population, discarding 53 to 60% of event cases: the short-window cases a deployed monitor will meet. The horizon-specific curve keeps more cases at short horizons but does not equalize pre-horizon alerting opportunity. Neither repairs the pooled headline median, and for the reason given in Section 2 the censoring is not independent of the event, with short- and long-run-up cases differing in surgery duration and case mix (Appendix A.12).

6 A task-specific reporting extension

Reporting conventions decide what interpretation is available to a reader. A clinician, policymaker or science writer meets a number and a name, not the scoring policy behind them, so what a standard asks for bounds what any of them can conclude. That is why auditing one metric is a communication result and not only a metrology one.

This section maps that case study onto what a general standard explicitly asks for. A TRIPOD+AI checklist was completed for this work by its own authors, before Sections 3 to 5 were conducted, and it did not request the disclosures that reveal the dependence. TRIPOD+AI is a general minimum **reporting** standard rather than a validity tool, and one author-completed checklist cannot assess a general framework in either direction. So what follows is a mapping, and the items in Section 6.2 are task-specific candidate disclosures that complement the standard rather than correct, invalidate or demonstrate non-adherence to it.

6.1 The items that were satisfied

Item 12e, performance measures and rationale, is where a lead time is declared: it asks *which* measures were used and *why*, not what else a measure could be responding to. **Item 23a, model performance**, asks for performance with uncertainty, and Table 1 satisfies it. So Table 1 is a compliant table in which a 24-minute lead sits beside a specificity of 0.406, with nothing in either item requiring a reader to put the two together. Four further items came close and missed (Table A8), and one pattern runs across them: the available pre-onset window distribution is never requested, and it is the variable the reported lead correlates with at $\rho = 0.501$ to 0.906.

Item 14, model fairness, is the near miss, and the most instructive item on the list. The checklist's own fairness answer records that arterial-line placement selects toward longer, higher-risk operations. The information that observation length is not distributed at random was therefore **already disclosed**, filed as a limitation, where it reads as a caveat about which patients the findings extend to. Nothing in the standard connects it to the timing metric three tables away.

6.2 Four candidate disclosures for continuously monitored models

Each is tied to a measurement in this paper, each is computable from artifacts a study of this kind already produces, and none requires a new experiment. They have not been validated as a general reporting framework.

1. **Report alert composition by elapsed observation time**: for every operating point, the share of alerts already active at the first eligible minute, those that fired within a short settling period, and those that fired later (Figure A2). One stacked bar would have made Section 4.1 visible on inspection.
2. **Report a window-conditional timing statistic as the primary timing quantity.** A horizon-specific detection curve may be added as a secondary disclosure, provided its denominator and its remaining dependence on pre-horizon alerting opportunity are stated. Any single median should be secondary and stratified by the distribution of available pre-onset windows it was measured over: the transformer reports 5 to 9 minutes for its shortest-window band and 46 to 59 for its longest, and the pooled 17 to 24 carries neither. Tables A6 and A7 show the two alternatives; Section 5.3 gives their limits.
3. **Report N(t) beside any event-aligned trajectory.** Here the risk set roughly halves between the event and 60 minutes before it (1,404 to 708, 2,804 to 1,433 and 2,641 to 1,374),

and the cases surviving to the far end are the long-window ones, the same selection that drives Section 3. **The denominator is not cosmetic**: recomputing the same bins on only the cases observable back to 60 minutes moves the first bin reaching AUROC 0.70 from 15 minutes in all three cohorts to 20, 35 and 20, so the horizon a reader would quote changes by up to twenty minutes on a reporting choice alone, and neither curve is the true one (Figure A3).

4. **Report a path-versus-displacement summary for any score read as a trend**, so that a series travelling much farther than its net displacement is qualified before it is described as rising or falling (Table A3 and Appendix A.4).

The first three are disclosure items in the ordinary TRIPOD sense, naming a quantity that must appear beside one already required; the fourth is closer to a diagnostic, and we offer it as the weakest of the four for that reason.

6.3 What this does and does not generalize to

The specific artifact does not generalize; its shape does. The shape is a quantity named for what a clinician would experience, computed from a window whose length varies across patients and is not reported, under a standard that requires the quantity and not the window. Any continuously read model evaluated with a first-crossing statistic over variable available windows is structurally susceptible to this dependence. Its empirical magnitude has to be measured in each setting, and the four items are the disclosure that would let a reader see it.

7 Limitations and scope

Retrospective, and one label. Every score is computed from cached data after the fact: a retrospective forward-running simulation, not deployment clearance. The outcome is a sustained excursion of mean arterial pressure below a fixed level, and both the level and the duration are conventions we did not vary; the label's dependence on a channel the models read is in Section 4.2.

One model fit, one family of operating points, and one draw of the control reference times. Models were trained at one seed and thresholds come from one validation split at one sensitivity target, transported unchanged; a different fit or target could move the composition of Section 5.1, the magnitudes of Section 3 and the operating points of Table 1, and we varied none of them. Control cases have no onset, so each is assigned a pseudo-onset sampled once at a fixed seed from the event cases' timing distribution. **Every control-side quantity here is conditional on that one draw**: the case-level specificity and positive predictive value of Table 1, the control-side minute share, and the cross-combination association of Section 4.1 that rests on them. Its resampling variability is not propagated into any interval we report. Redrawing the reference times would require rerunning the complete control-side evaluation pipeline for every model and cohort, so we report all control-side quantities as conditional on the single pre-specified draw rather than presenting an unperformed sensitivity analysis.

Properties of a pipeline, not of an architecture. Section 4.1 describes what one model does under one scoring policy on these cohorts, not what sequence models must do. Section 3's effect is present in all five families, the ratio running 0.455 to 0.944, with the transformer the limiting case rather than the exception.

The references bias upward. Both shuffle variants make agreement easier to obtain, in the direction that favors our own reading, and they change the detected-case set. They bound the dependence rather than estimating a case-independent process.

Communication and reporting claims, and one clinical domain. We analyze the interpretation invited by the metric's name, not measured clinician comprehension. The TRIPOD+AI assessment was author-completed, and the four disclosures were demonstrated here rather than validated across reporting settings or stakeholder groups. Three cohorts, two health systems and two countries still give one procedure setting and one monitored variable, so they cannot settle whether Section 6.2's four items are the right four elsewhere.

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Appendix A Methods and complete per-cell results

A.1 Cohorts, models, training and compute

Cohorts and eligibility. Two public datasets supply the three cohorts: VitalDB, and MOVER, whose single release carries both an EPIC-derived and an SIS-derived arm. Both are de-identified and both are obtained under their providers' data use agreements. Development used MOVER's EPIC arm. The subject-aware split at seed 1004 gave 6,409 training, 1,375 validation and 1,404 test cases out of 9,188, split on unique subject identifiers so that a patient with several operations lands in one fold. The test split holds 1,404 cases, 786 with a sustained event, 1,145 subjects. External evaluation used VitalDB, with 2,804 cases, 1,007 events and 2,730 subjects, and the SIS-derived arm of MOVER, with 2,641 cases and 1,409 events, neither of which contributed to training, calibration or threshold selection. Per-minute eligibility drops the induction prefix, 25 minutes where the record is anchored on anesthesia start, 10 where it is anchored on surgical start, and every minute at or after first onset, so no model is asked to predict an event already in progress. Event cases required five valid pre-onset minutes after that exclusion and controls thirty. Procedure group is never a model input in any family: it separates the cohorts almost perfectly and would be learned as cohort identity.

Features. The same twenty inputs for every family: sixteen per-minute channels, being MAP, systolic and diastolic pressure, their non-invasive counterparts, heart rate, SpO2, end-tidal CO2, temperature, FiO2, MAC, tidal volume, respiratory rate, PEEP and peak inspiratory pressure, and four demographics, being age, sex, height and weight. The families differ in how they consume them. The transformer takes each non-missing cell as a (type, time, value) triplet with demographics as a fixed prefix at time zero, up to 2,048 tokens over a 120-minute window, so missing cells produce no token and are never imputed. The LSTM takes the normalized per-minute window as a value-and-mask pair per channel. Logistic regression, random forest and XGBoost take the last observation carried forward per channel plus the demographics. Per-channel median and interquartile scaling was fitted on the training cases alone and applied unchanged everywhere else.

Architectures and training. Seed 1004 throughout. The transformer is a STraTS-style encoder, a type embedding plus a value MLP plus a time MLP summed per triplet, two encoder layers, model width 128, four heads, feed-forward width 512, dropout 0.1, a learnable classification token and a two-class softmax head; AdamW at learning rate 1.2e-3, weight decay 1e-2 applied to two-dimensional tensors only, gradient clipping at 1.0, a cosine schedule with 5% warmup, batch size 64, cross-entropy, up to 50 epochs with early stopping on validation AUROC at patience 10, under bfloat16 autocast. The LSTM is one layer of width 128 with dropout 0.1, AdamW at 1e-3 with weight decay 1e-2, batch size 64, up to 40 epochs at patience 8. Logistic regression is median imputation and standardization followed by an L2 fit at up to 2,000 iterations. The random forest is 400 trees. XGBoost is 600 rounds at depth 6, learning rate 0.05, row and column subsampling at 0.8, histogram tree method, early stopping after 40 rounds on validation AUC. No hyperparameter search was run over any of these values.

Compute. Every model was trained on one Apple M3 Ultra desktop with 512 GB of unified memory, using PyTorch 2.10 on the Metal backend with bfloat16 autocast; no GPU cluster was used. Wall clock for the five models reported here was 40.3 minutes for the transformer, 14.5 for the LSTM, 3.6 for the random forest, and 34 and 33 seconds for XGBoost and logistic regression. The project trained more than this paper reports: each family was also trained at seven lead-time offsets, of which the transformer arm alone took 3.8 hours. The analyses in this paper add no training and read cached per-minute scores on the same machine. Software was PyTorch 2.10.0, scikit-learn 1.7.1, XGBoost 3.3.0, NumPy 2.2.6, pandas 2.3.2 and SciPy 1.16.1.

Data and code. Both datasets are obtained from their own project sites once access is granted: VitalDB under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International license and a data use agreement, MOVER from the UCI Machine Learning Repository under a data use agreement, one download supplying both arms used here. Each original collection was approved by its own institutional review board with consent waived after de-identification. The exemption from review and the consent waiver for the present secondary analysis were granted by an institutional review board whose identity and protocol number are withheld for anonymous review. This work added no data collection and no patient contact. Analysis code is available at an anonymized repository, `https://anonymous.4open.science/r/ioh-leadtime-anon-BD0E`, under the Apache License 2.0. Neither dataset may be redistributed under the agreements above, so the repository car-

ries the analysis code and the aggregate result files behind the tables and figures, and no patient-level data of any kind.

A.2 Lead time by quantile-derived band of observed window, all fifteen cells

Table A1 | Median alert lead time by quantile-derived band of observed window, all fifteen cells. Main text Section 3.1 reports the ITE Transformer. The gradient is positive in every cell. `Bands` is the number of quantile-derived groups actually realized: where the observed window is censored at the 60-minute alert window the upper quintiles collapse onto a single value, and duplicate edges are dropped rather than bands invented. The gradient is the difference of the highest and lowest band medians, and its interval is the paired cluster bootstrap of that difference: one weight matrix is drawn per cell and reused across bands, and the band edges and the window span are held fixed at their full-sample values so the ratio's denominator is not itself resampled. The interval excludes zero in all fifteen cells.

Model	Cohort	Bands	Window Q1→last, min	Lead Q1→last, min	Gradient, min (95% CI)	Lead min per window min (95% CI)
Transformer	MOVER (EPIC) test	4	10 → 60	9 → 46	+37 [+35, +41]	0.74 [0.70, 0.82]
Transformer	VitalDB	5	7 → 60	5 → 59	+54 [+51, +55]	1.02 [0.96, 1.04]
Transformer	MOVER (SIS)	4	8 → 60	6 → 46	+40 [+37, +43]	0.77 [0.71, 0.83]
LSTM	MOVER (EPIC) test	4	10 → 60	6 → 44	+38 [+35, +41]	0.77 [0.70, 0.82]
LSTM	VitalDB	5	6 → 60	3 → 34	+31 [+21, +48]	0.57 [0.39, 0.89]
LSTM	MOVER (SIS)	4	9 → 60	5 → 31	+26 [+19, +30]	0.51 [0.37, 0.59]
XGBoost	MOVER (EPIC) test	3	12 → 60	8 → 38	+30 [+26, +34]	0.62 [0.54, 0.71]
XGBoost	VitalDB	3	9 → 60	3 → 30	+27 [+22, +30]	0.53 [0.43, 0.58]
XGBoost	MOVER (SIS)	3	12 → 60	5 → 36	+31 [+28, +33]	0.65 [0.58, 0.69]
Random forest	MOVER (EPIC) test	3	13 → 60	8 → 38	+30 [+26, +34]	0.64 [0.55, 0.72]
Random forest	VitalDB	3	9 → 60	3 → 34	+31 [+29, +34]	0.61 [0.57, 0.67]
Random forest	MOVER (SIS)	3	12 → 60	6 → 40	+34 [+31, +36]	0.71 [0.65, 0.75]
Logistic reg.	MOVER (EPIC) test	3	11 → 60	8 → 40	+32 [+28, +35]	0.65 [0.57, 0.71]
Logistic reg.	VitalDB	3	10 → 60	3 → 28	+26 [+20, +28]	0.51 [0.40, 0.56]
Logistic reg.	MOVER (SIS)	3	12 → 60	5 → 38	+33 [+29, +36]	0.69 [0.61, 0.75]

A.3 Both nulls, all fifteen cells

Table A2 | Both nulls, all fifteen cells. Observed ratio against the uniform-placement null of Section 3.3 and both variants of the cross-case shuffle. The uniform null's expectation is 0.5 by construction; the two shuffle variants differ only in how a donor shorter than its recipient is handled, and both bias the null upward. Intervals are the 2.5th and 97.5th percentiles of the null distribution.

Model	Cohort	Observed ratio	Uniform null	Shuffle null, hold	Shuffle null, length-matched	Observed ρ	Shuffle ρ, hold	Shuffle ρ, length-matched
Transformer	MOVER (EPIC) test	0.944	0.524 [0.491, 0.567]	0.970 [0.967, 0.974]	0.962 [0.951, 0.971]	0.706	0.888 [0.855, 0.920]	0.783 [0.705, 0.856]
Transformer	VitalDB	0.923	0.532 [0.500, 0.573]	0.966 [0.963, 0.967]	0.957 [0.947, 0.969]	0.906	0.941 [0.926, 0.955]	0.933 [0.890, 0.967]
Transformer	MOVER (SIS)	0.899	0.527 [0.500, 0.567]	0.949 [0.941, 0.950]	0.918 [0.900, 0.938]	0.801	0.877 [0.855, 0.895]	0.775 [0.713, 0.827]
LSTM	MOVER (EPIC) test	0.851	0.522 [0.483, 0.563]	0.913 [0.900, 0.923]	0.866 [0.833, 0.893]	0.650	0.825 [0.784, 0.860]	0.673 [0.596, 0.743]
LSTM	VitalDB	0.619	0.534 [0.500, 0.583]	0.870 [0.850, 0.883]	0.752 [0.667, 0.833]	0.759	0.845 [0.798, 0.886]	0.705 [0.593, 0.804]
LSTM	MOVER (SIS)	0.717	0.524 [0.500, 0.562]	0.873 [0.857, 0.887]	0.700 [0.633, 0.767]	0.501	0.682 [0.636, 0.727]	0.455 [0.368, 0.547]
XGBoost	MOVER (EPIC) test	0.719	0.519 [0.483, 0.558]	0.842 [0.814, 0.867]	0.760 [0.705, 0.800]	0.558	0.701 [0.651, 0.751]	0.581 [0.503, 0.649]
XGBoost	VitalDB	0.467	0.521 [0.483, 0.556]	0.723 [0.667, 0.768]	0.591 [0.517, 0.652]	0.599	0.636 [0.569, 0.698]	0.493 [0.400, 0.580]
XGBoost	MOVER (SIS)	0.600	0.518 [0.500, 0.550]	0.779 [0.750, 0.800]	0.681 [0.633, 0.722]	0.573	0.632 [0.590, 0.673]	0.488 [0.429, 0.541]
Random forest	MOVER (EPIC) test	0.714	0.519 [0.483, 0.556]	0.839 [0.816, 0.867]	0.777 [0.733, 0.817]	0.584	0.714 [0.665, 0.764]	0.599 [0.531, 0.667]
Random forest	VitalDB	0.533	0.523 [0.500, 0.556]	0.764 [0.721, 0.800]	0.685 [0.625, 0.733]	0.651	0.678 [0.622, 0.729]	0.555 [0.479, 0.627]
Random forest	MOVER (SIS)	0.667	0.518 [0.500, 0.550]	0.811 [0.783, 0.833]	0.741 [0.700, 0.783]	0.618	0.659 [0.625, 0.692]	0.536 [0.486, 0.589]
Logistic reg.	MOVER (EPIC) test	0.781	0.520 [0.483, 0.557]	0.872 [0.849, 0.898]	0.802 [0.767, 0.842]	0.597	0.739 [0.694, 0.779]	0.612 [0.544, 0.676]
Logistic reg.	VitalDB	0.455	0.521 [0.471, 0.567]	0.721 [0.667, 0.781]	0.577 [0.500, 0.650]	0.607	0.642 [0.549, 0.720]	0.486 [0.374, 0.584]
Logistic reg.	MOVER (SIS)	0.618	0.518 [0.500, 0.550]	0.782 [0.750, 0.800]	0.734 [0.692, 0.767]	0.597	0.644 [0.607, 0.682]	0.548 [0.495, 0.595]

Three cells in this table are the ones Section 3.3 names as not exceeding the uniform reference, all of them VitalDB. The observed ratio sits below it for XGBoost by -0.054 and for logistic regression by -0.067, and random forest is indistinguishable from it (+0.011, $P = 0.824$). For the transformer the excess over the uniform reference is +0.372 to +0.421. Under the length-matched shuffle variant the transformer's null spans 0.77 to 0.93.

A.4 Path decomposition

Table A3 | Path decomposition over the window 60 to 5 minutes before onset. Event episodes have observed scores at both window edges. Directedness is net displacement divided by total distance traveled, so 1 is a monotone climb and 0 is a round trip. Path per net is total distance divided by the absolute net change. Episodes observed at every bin center in the window; median 11 steps per episode. Intervals are 95% cluster bootstrap. Only this window is reported: over 60 to 10 minutes the transformer's net displacement is indistinguishable from zero in two cohorts, and the ratio is then a division by approximately nothing.

Cohort	Model	Episodes	Directedness (95% CI)	Path per unit net
MOVER (EPIC) test	Transformer	313	0.025 [0.002, 0.050]	40.7
MOVER (EPIC) test	LSTM	313	0.043 [0.024, 0.057]	25.9
MOVER (EPIC) test	XGBoost	313	0.049 [0.033, 0.070]	20.0
MOVER (EPIC) test	Random forest	313	0.067 [0.038, 0.093]	16.3
MOVER (EPIC) test	Logistic reg.	313	0.061 [0.044, 0.089]	17.6
VitalDB	Transformer	433	0.021 [0.005, 0.038]	146.6
VitalDB	LSTM	433	0.059 [0.035, 0.070]	28.2
VitalDB	XGBoost	433	0.069 [0.049, 0.099]	17.9
VitalDB	Random forest	433	0.075 [0.053, 0.094]	14.4
VitalDB	Logistic reg.	433	0.088 [0.063, 0.108]	12.6
MOVER (SIS)	Transformer	670	0.089 [0.066, 0.122]	13.1
MOVER (SIS)	LSTM	670	0.142 [0.118, 0.183]	7.9
MOVER (SIS)	XGBoost	670	0.144 [0.120, 0.167]	7.5
MOVER (SIS)	Random forest	670	0.151 [0.132, 0.168]	7.0
MOVER (SIS)	Logistic reg.	670	0.134 [0.119, 0.154]	7.9

Directedness is net displacement divided by total distance traveled: over the window from 60 to 5 minutes before onset it is **+0.021 to +0.151 across all fifteen cells**, every interval excluding zero, so no family approaches a directed climb. **Directedness describes path shape and does not establish when or whether a particular threshold is touched.** Three further things belong here. The ratio of the median path length to the absolute median net displacement is 7.0 to 146.6 across the fifteen cells; because that ratio becomes unstable as net displacement approaches zero, directedness is the primary summary and both quantities describe path shape rather than prove a crossing mechanism. The transformer sits at the volatile end of both (directedness +0.021 to +0.089, path per unit net displacement 13.1 to 146.6) and logistic regression at the least volatile (+0.061 to +0.134, 7.9 to 17.6). And averaged over cases the score still behaves as a reader expects, rising where an event follows and staying much flatter where none does, but the transformer's event and control means are indistinguishable an hour before onset where the tabular families' are already apart (Figure A4). The property is present under this task and scoring policy and the transformer is its limiting case; the design does not isolate architecture from representation or training.

A.5 Case-level operating point, complete detail

Table A4 | Case-level operating point, all fifteen cells. Main text Table 1 carries the ITE Transformer's three rows without intervals; this is every family and every interval, and it is where the cross-combination claim of Section 4.1 is checked. Sensitivity and specificity are case-level as defined in Section 2, NNE is alerting cases per detected event and equals 1/PPV, and `Min. above thr.` is the median share of an event case's monitored minutes at or above the transported threshold. Cohort composition, prevalence, PPV in excess of prevalence, the control-side minute share and the share of event cases already alerting by the first elapsed minute are Table A10. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS).

Model	Cohort	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NNE (95% CI)	Min. above thr., events (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	0.777 [0.746, 0.806]	0.406 [0.368, 0.443]	0.625 [0.594, 0.656]	1.60 [1.53, 1.68]	0.523 [0.444, 0.575]	24 [21, 27]
ITE Transformer	VitalDB	0.679 [0.651, 0.707]	0.379 [0.358, 0.403]	0.380 [0.357, 0.402]	2.63 [2.49, 2.80]	0.444 [0.389, 0.500]	17 [15, 19]
ITE Transformer	MOVER (SIS)	0.657 [0.632, 0.682]	0.467 [0.441, 0.494]	0.585 [0.561, 0.609]	1.71 [1.64, 1.78]	0.200 [0.150, 0.238]	22 [20, 24]
LSTM	MOVER (EPIC) test	0.807 [0.779, 0.834]	0.474 [0.433, 0.513]	0.661 [0.631, 0.690]	1.51 [1.45, 1.58]	0.267 [0.209, 0.308]	23 [20, 25]
LSTM	VitalDB	0.492 [0.460, 0.523]	0.812 [0.795, 0.830]	0.595 [0.563, 0.629]	1.68 [1.59, 1.78]	0.000 [0.000, 0.017]	9 [8, 10]
LSTM	MOVER (SIS)	0.551 [0.526, 0.577]	0.709 [0.683, 0.735]	0.685 [0.657, 0.712]	1.46 [1.40, 1.52]	0.017 [0.017, 0.020]	15 [14, 18]
XGBoost	MOVER (EPIC) test	0.810 [0.782, 0.835]	0.608 [0.570, 0.647]	0.725 [0.696, 0.754]	1.38 [1.33, 1.44]	0.117 [0.100, 0.133]	21 [18, 23]
XGBoost	VitalDB	0.694 [0.665, 0.721]	0.754 [0.735, 0.774]	0.613 [0.584, 0.640]	1.63 [1.56, 1.71]	0.067 [0.050, 0.067]	13 [12, 16]
XGBoost	MOVER (SIS)	0.790 [0.768, 0.810]	0.638 [0.610, 0.666]	0.714 [0.691, 0.736]	1.40 [1.36, 1.45]	0.100 [0.100, 0.117]	19 [18, 22]
Random forest	MOVER (EPIC) test	0.828 [0.801, 0.854]	0.555 [0.515, 0.597]	0.703 [0.674, 0.732]	1.42 [1.37, 1.48]	0.150 [0.133, 0.167]	22 [20, 24]
Random forest	VitalDB	0.803 [0.778, 0.827]	0.674 [0.651, 0.695]	0.580 [0.553, 0.604]	1.72 [1.65, 1.81]	0.133 [0.118, 0.150]	15 [12, 18]
Random forest	MOVER (SIS)	0.849 [0.830, 0.867]	0.559 [0.532, 0.586]	0.688 [0.666, 0.710]	1.45 [1.41, 1.50]	0.172 [0.167, 0.183]	22 [20, 25]
Logistic regression	MOVER (EPIC) test	0.837 [0.810, 0.864]	0.518 [0.477, 0.557]	0.688 [0.658, 0.719]	1.45 [1.39, 1.52]	0.139 [0.114, 0.162]	23 [21, 26]
Logistic regression	VitalDB	0.461 [0.431, 0.492]	0.825 [0.808, 0.842]	0.596 [0.565, 0.631]	1.68 [1.59, 1.77]	0.000 [0.000, 0.000]	14 [10, 16]
Logistic regression	MOVER (SIS)	0.790 [0.769, 0.812]	0.585 [0.559, 0.611]	0.685 [0.662, 0.708]	1.46 [1.41, 1.51]	0.117 [0.100, 0.130]	21 [19, 23]

A.6 Detection rate by band of observed window, all fifteen cells

Table A5 | Detection rate by band of observed pre-onset window, all fifteen cells. Bands are quantiles of the observed window cut over **all** event cases with a positive observed window, not

over the detected subset Table A9 and Table A1 cut on, because a within-band detection rate needs every event case of that window length in its denominator. The two banding schemes are therefore not interchangeable, the band counts here are event-case counts rather than detected counts, and the median leads differ slightly from Table A1's. The detection rate summed over bands reproduces the case-level sensitivity of Table 1 exactly in all fifteen cells, which is the check that the denominators are the same event cases. Intervals are 95% cluster bootstrap.

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
ITE Transformer	MOVER (EPIC) test	1	159	159	11	1.000 [1.000, 1.000]	9 [8, 10]
ITE Transformer	MOVER (EPIC) test	2	164	164	23	1.000 [1.000, 1.000]	22 [20, 24]
ITE Transformer	MOVER (EPIC) test	3	150	149	44	0.993 [0.979, 1.000]	42 [40, 43]
ITE Transformer	MOVER (EPIC) test	4	313	139	60	0.444 [0.388, 0.498]	54 [33, 58]
ITE Transformer	VitalDB	1	204	204	8	1.000 [1.000, 1.000]	6 [6, 7]
ITE Transformer	VitalDB	2	203	203	18	1.000 [1.000, 1.000]	16 [14, 17]
ITE Transformer	VitalDB	3	600	277	60	0.462 [0.421, 0.500]	44 [42, 46]
ITE Transformer	MOVER (SIS)	1	287	283	10	0.986 [0.971, 0.997]	7 [7, 8]
ITE Transformer	MOVER (SIS)	2	285	281	25	0.986 [0.971, 0.997]	23 [22, 24]
ITE Transformer	MOVER (SIS)	3	837	362	60	0.432 [0.398, 0.465]	46 [43, 49]
LSTM	MOVER (EPIC) test	1	159	148	11	0.931 [0.889, 0.970]	7 [6, 8]
LSTM	MOVER (EPIC) test	2	164	157	23	0.957 [0.925, 0.987]	20 [18, 21]
LSTM	MOVER (EPIC) test	3	150	141	44	0.940 [0.897, 0.974]	39 [37, 41]
LSTM	MOVER (EPIC) test	4	313	188	60	0.601 [0.545, 0.652]	40 [34, 46]
LSTM	VitalDB	1	204	166	8	0.814 [0.760, 0.868]	3 [3, 4]
LSTM	VitalDB	2	203	147	18	0.724 [0.660, 0.783]	10 [9, 12]
LSTM	VitalDB	3	600	182	60	0.303 [0.268, 0.340]	32 [28, 34]
LSTM	MOVER (SIS)	1	287	196	10	0.683 [0.628, 0.738]	6 [5, 6]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
LSTM	MOVER (SIS)	2	285	188	25	0.660 [0.604, 0.714]	18 [16, 21]
LSTM	MOVER (SIS)	3	837	393	60	0.470 [0.437, 0.504]	33 [29, 36]
XGBoost	MOVER (EPIC) test	1	159	110	11	0.692 [0.619, 0.760]	7 [5, 9]
XGBoost	MOVER (EPIC) test	2	164	132	23	0.805 [0.743, 0.865]	17 [16, 19]
XGBoost	MOVER (EPIC) test	3	150	130	44	0.867 [0.806, 0.918]	32 [28, 36]
XGBoost	MOVER (EPIC) test	4	313	265	60	0.847 [0.806, 0.885]	40 [34, 44]
XGBoost	VitalDB	1	204	110	8	0.539 [0.471, 0.606]	3 [2, 3]
XGBoost	VitalDB	2	203	132	18	0.650 [0.587, 0.718]	7 [5, 9]
XGBoost	VitalDB	3	600	457	60	0.762 [0.728, 0.794]	26 [23, 30]
XGBoost	MOVER (SIS)	1	287	179	10	0.624 [0.570, 0.678]	5 [4, 5]
XGBoost	MOVER (SIS)	2	285	228	25	0.800 [0.753, 0.844]	14 [13, 16]
XGBoost	MOVER (SIS)	3	837	706	60	0.843 [0.819, 0.866]	35 [33, 37]
Random forest	MOVER (EPIC) test	1	159	107	11	0.673 [0.596, 0.746]	6 [5, 8]
Random forest	MOVER (EPIC) test	2	164	135	23	0.823 [0.762, 0.880]	17 [15, 18]
Random forest	MOVER (EPIC) test	3	150	129	44	0.860 [0.800, 0.914]	30 [27, 34]
Random forest	MOVER (EPIC) test	4	313	280	60	0.895 [0.861, 0.928]	41 [35, 45]
Random forest	VitalDB	1	204	140	8	0.686 [0.626, 0.747]	3 [2, 3]
Random forest	VitalDB	2	203	157	18	0.773 [0.711, 0.831]	7 [6, 10]
Random forest	VitalDB	3	600	512	60	0.853 [0.825, 0.882]	32 [30, 35]
Random forest	MOVER (SIS)	1	287	201	10	0.700 [0.646, 0.752]	5 [5, 6]
Random forest	MOVER (SIS)	2	285	245	25	0.860 [0.819, 0.899]	16 [14, 17]

Model	Cohort	Band	Event cases	Detected	Window median, min	Detection rate (95% CI)	Median lead, min (95% CI)
Random forest	MOVER (SIS)	3	837	750	60	0.896 [0.875, 0.916]	38 [36, 41]
Logistic regression	MOVER (EPIC) test	1	159	119	11	0.748 [0.678, 0.814]	7 [6, 9]
Logistic regression	MOVER (EPIC) test	2	164	138	23	0.841 [0.787, 0.896]	18 [17, 20]
Logistic regression	MOVER (EPIC) test	3	150	134	44	0.893 [0.843, 0.940]	36 [32, 38]
Logistic regression	MOVER (EPIC) test	4	313	267	60	0.853 [0.812, 0.892]	42 [36, 46]
Logistic regression	VitalDB	1	204	71	8	0.348 [0.283, 0.412]	3 [2, 3]
Logistic regression	VitalDB	2	203	95	18	0.468 [0.404, 0.539]	7 [6, 8]
Logistic regression	VitalDB	3	600	298	60	0.497 [0.454, 0.537]	26 [22, 30]
Logistic regression	MOVER (SIS)	1	287	178	10	0.620 [0.566, 0.676]	5 [5, 5]
Logistic regression	MOVER (SIS)	2	285	221	25	0.775 [0.727, 0.821]	14 [12, 16]
Logistic regression	MOVER (SIS)	3	837	714	60	0.853 [0.829, 0.876]	36 [34, 38]

A.7 Replacement estimands, all horizons

Table A6 | Fixed-horizon detection rate at all seven horizons, and time to first alert. The fixed-horizon detection rate is the share of event cases observed for at least the stated number of minutes before onset whose first alert came at least that many minutes before onset. Point estimates are shown here; each is a share of the cases in its own column, and its 95% cluster bootstrap interval is computed exactly as elsewhere in this appendix. Section 5.3 quotes the range across cells at the 10- and 30-minute horizons. `n observed` is the denominator at the first and last horizon, and its fall is the reason a single median is not comparable across cohorts. The 60-minute column is the share of fully observed cases that alerted at the very first opportunity.

Model	Cohort	5 min	10 min	15 min	20 min	30 min	45 min	60 min	n observed $\geq 5 \rightarrow \geq 60$
ITE Transformer	MOVER (EPIC) test	0.733	0.690	0.646	0.605	0.512	0.339	0.170	786 → 311
ITE Transformer	VitalDB	0.638	0.560	0.509	0.463	0.390	0.261	0.154	1007 → 429
ITE Transformer	MOVER (SIS)	0.594	0.547	0.504	0.469	0.377	0.247	0.125	1409 → 665
LSTM	MOVER (EPIC) test	0.721	0.663	0.635	0.597	0.490	0.331	0.109	786 → 311
LSTM	VitalDB	0.343	0.278	0.265	0.223	0.173	0.102	0.030	1007 → 429
LSTM	MOVER (SIS)	0.453	0.392	0.346	0.319	0.253	0.167	0.047	1409 → 665
XGBoost	MOVER (EPIC) test	0.688	0.651	0.615	0.575	0.484	0.367	0.119	786 → 311
XGBoost	VitalDB	0.502	0.470	0.453	0.425	0.366	0.237	0.054	1007 → 429
XGBoost	MOVER (SIS)	0.656	0.597	0.565	0.520	0.460	0.326	0.080	1409 → 665
Random forest	MOVER (EPIC) test	0.709	0.674	0.638	0.611	0.506	0.393	0.138	786 → 311
Random forest	VitalDB	0.589	0.559	0.559	0.530	0.478	0.347	0.103	1007 → 429
Random forest	MOVER (SIS)	0.736	0.674	0.640	0.600	0.526	0.403	0.114	1409 → 665
Logistic regression	MOVER (EPIC) test	0.726	0.692	0.658	0.630	0.547	0.401	0.122	786 → 311
Logistic regression	VitalDB	0.338	0.307	0.303	0.274	0.233	0.137	0.026	1007 → 429
Logistic regression	MOVER (SIS)	0.664	0.611	0.573	0.546	0.474	0.353	0.093	1409 → 665

A.8 The full alert window as a constant

Table A7 | Event cases observed for the whole 60-minute alert window. Restricting to these cases makes the observation window a constant rather than a covariate, which is the window-conditional estimand of Section 5.3. `n` is the number of event cases observed for the whole window and `Detected` the share of them that alert at all. `Alerted by min 1` is one minus the Kaplan-Meier survival of time to first alert at the first elapsed monitoring minute, with each case censored at the end of its own window; it is the censoring-aware form of Section 4.1's settling. The censoring is not independent of the event, because for an event case a short observable run-up is an early-onset event rather than missing data, so this describes what the rule did under the observation these cohorts supplied. Intervals are 95% cluster bootstrap.

Model	Cohort	n	Detected	Median lead, min (95% CI)	Alerted by min 1 (95% CI)
ITE Transformer	MOVER (EPIC) test	311	0.441	47 [33, 59]	0.532 [0.497, 0.565]
ITE Transformer	VitalDB	429	0.247	60 [60, 60]	0.460 [0.428, 0.491]
ITE Transformer	MOVER (SIS)	665	0.304	57 [51, 59]	0.308 [0.285, 0.332]
LSTM	MOVER (EPIC) test	311	0.598	40 [34, 46]	0.228 [0.198, 0.257]
LSTM	VitalDB	429	0.200	33 [21, 45]	0.087 [0.071, 0.105]
LSTM	MOVER (SIS)	665	0.433	28 [22, 33]	0.138 [0.120, 0.157]
XGBoost	MOVER (EPIC) test	311	0.846	40 [34, 44]	0.188 [0.161, 0.215]
XGBoost	VitalDB	429	0.762	33 [30, 36]	0.062 [0.047, 0.077]
XGBoost	MOVER (SIS)	665	0.854	37 [34, 41]	0.119 [0.101, 0.136]
Random forest	MOVER (EPIC) test	311	0.894	41 [35, 45]	0.183 [0.157, 0.212]
Random forest	VitalDB	429	0.851	40 [36, 44]	0.101 [0.082, 0.121]
Random forest	MOVER (SIS)	665	0.910	42 [39, 45]	0.166 [0.147, 0.186]
Logistic regression	MOVER (EPIC) test	311	0.852	42 [36, 45]	0.226 [0.198, 0.255]
Logistic regression	VitalDB	429	0.490	31 [26, 37]	0.036 [0.025, 0.048]
Logistic regression	MOVER (SIS)	665	0.857	40 [37, 43]	0.121 [0.104, 0.138]

A.9 The reporting items that came close

Table A8 | The four TRIPOD+AI items that came close and missed. Main text Section 6.1 keeps items 12e, 23a and 14. The item numbering is that of the TRIPOD+AI statement [Collins et al., 2024]. This is an author-completed self-assessment of one study, not a validity assessment of the standard.

Item	What it asked for	What it did not ask for
12a, how the model was applied in validation	Calibration and thresholds fitted on the development validation split and transported unchanged, which is reported	Anything about what that disciplined choice implies; it is the exact setting in which Section 3 holds, so doing the right thing here does not help
20b, participant characteristics	Participant characteristics, which are reported	The distribution of the observed pre-onset window, the variable the reported lead correlates with at ρ = 0.501 to 0.906
21, event counts	Event and non-event counts, which are reported	How long each case was observed before its event
8c, blinding of outcome assessment	A blinding statement; answered "not applicable" because the outcome is computed from a monitored channel that is also a predictor	The closest the checklist comes to Section 4, and still about blinding rather than about what a timing metric measures
15, model output	The model output and its threshold; answered with the transported threshold itself	Anything about the operating point that threshold defines, which is where the artifact lives (Table 1)

A.10 The relationship in all five families

Figure 1 plots the transformer in three cohorts. The lead-to-window relationship it shows is not particular to that model: in the per-case scatter for MOVER (EPIC) test it is present in every family, with the transformer the limiting case rather than the exception. Figure A1 covers that cohort only; the other two are in Table A1, where the gradient is positive with its interval excluding zero in every cell.

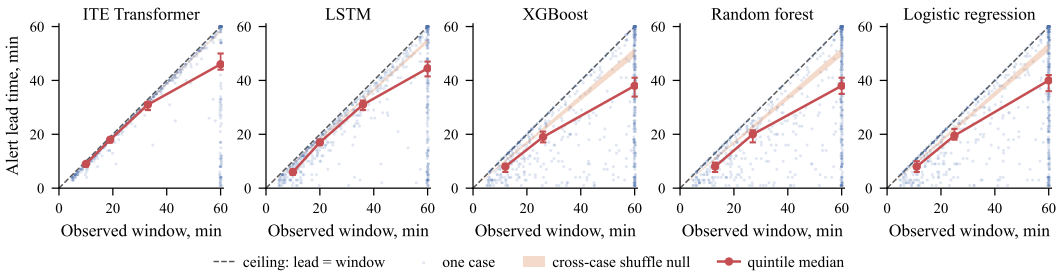


Figure A1: **Alert lead time against observed window length, five model families, MOVER (EPIC) test.** Axes and overlays as in Figure 1. The relationship holds in every family; the transformer is the limiting case rather than the exception.

A.11 Alert composition

Section 5.1 reports the prevalent share as a single number per cell. For the transformer that share is 8.7, 9.6 and 9.0% in the three cohorts, and removing those cases moves its median lead from 24, 17 and 22 minutes to 21, 15 and 19. Splitting detected event cases by when their first crossing fell separates alerts that were already active at the first eligible minute from those raised while the score was still settling and those raised later, which is the disclosure Section 6.2's first item asks for.

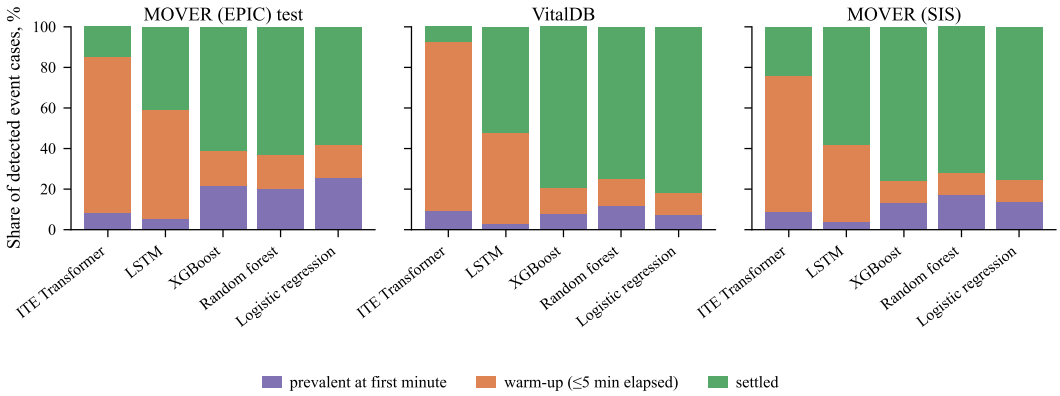


Figure A2: **Composition of detected event cases by elapsed monitoring time at the first crossing.** Prevalent cases are above threshold at their first eligible minute; warm-up cases cross within five minutes of it; settled cases cross later. Five model families, three cohorts, at the transported 80%-sensitivity operating point. The composition differs sharply across families, in that the transformer's alerts are concentrated in the first five minutes, the LSTM's are partly so and the tabular families' are not, while the lead-to-window relationship of Section 3 holds in all of them. That is why Section 4 separates the transformer's direct mechanism from the cross-family directedness result.

A.12 Risk-set composition, and what fixing the cohort does

The population trajectory scores every case that has a minute in the bin, so its denominator shrinks and its composition shifts as the bin moves away from onset. For an event case the observable run-up *is* its time to onset, so a short run-up is not missing data. It is an early-onset event, and the far bins keep only the long run-ups. That is the same selection that drives Section 3.

Fixing the cohort moves the trajectory in both directions (Figure A3). Recomputing the same bins on the cases observable back to 60 minutes raises the far end and lowers the near end, and the two curves cross at 16.4, 12.6 and 14.9 minutes before onset. The first bin reaching AUROC 0.70 moves from 15 minutes in all three cohorts to 20, 35 and 20. Neither curve is the true one: which

denominator is used is a reporting choice, it changes the horizon a reader would quote by up to twenty minutes, and it is not currently disclosed. That is the whole of Section 6.2's third item.

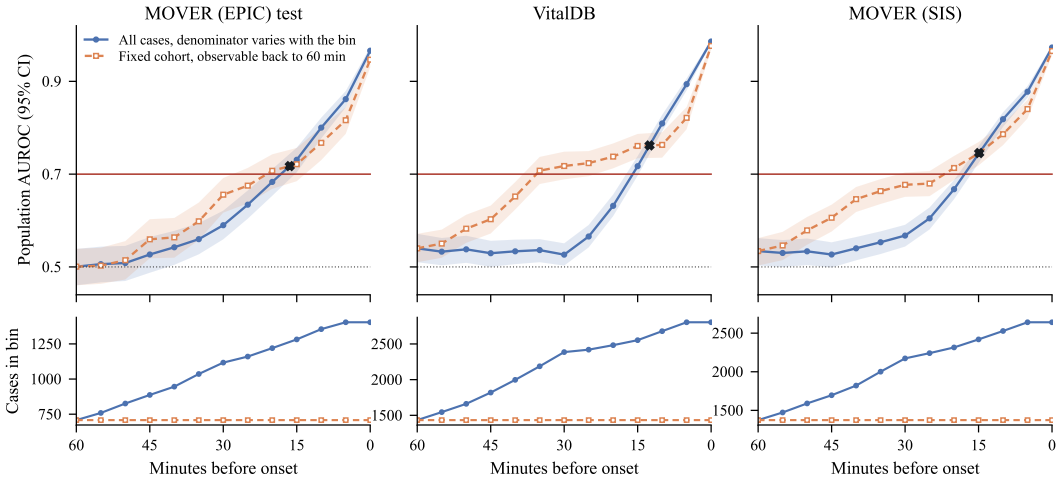


Figure A3: **Population AUROC with its own risk set, event-aligned.** ITE Transformer, 2-minute bins from 60 minutes before onset to onset, in three held-out cohorts. Solid and filled is every case with a minute in the bin, the convention in the published literature, whose denominator varies; dashed and open is the cohort fixed to cases observable back to 60 minutes. Bands are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). The lower row is the number of cases contributing to each bin, on its own scale per cohort. The dotted line marks 0.5, the solid rule 0.70, and the cross marks where the two curves meet.

A.13 Event and control score on the event-aligned axis

The shape a reader expects of a continuously read score, rising in the cases that go on to have an event and much flatter in the rest, is **present in all fifteen cells** (Figure A4). Three things about it are worth reporting, and none is visible in a trend statistic.

When the separation arrives runs opposite to near-onset discrimination. For the three tabular families the event and control means are already distinguishable 60 minutes before onset in every cohort, the gap at that range running +0.049 to +0.109 with every interval excluding zero. For the transformer the gap at 60 minutes **covers zero in all three cohorts** (-0.023 [-0.080, 0.035], -0.014 [-0.057, 0.030] and -0.006 [-0.043, 0.032]), and first excludes it at 35, 20 and 30 minutes. The LSTM sits between the two. The family whose scores separate latest is the one with the highest discrimination near onset.

The event mean rises toward onset, and for the transformer the control mean falls by as much. Over the hour the transformer's event mean rises +0.239, +0.266 and +0.269, which is the tendency a reader expects and it is substantial, while its control mean falls -0.252, -0.263 and -0.196, so about half of the widening is carried by the controls. The LSTM's control mean also falls, by 0.059 to 0.185. For the tabular families the control mean moves between -0.070 and +0.021, rising in several cells, and the widening is almost entirely event-side. So "the score rises in patients who are about to have an event" holds in every cell here; what it leaves out is that for the transformer an equal share of the widening is the control mean coming down, and an event-only trend statistic cannot separate the two.

The split follows the representation more than the architecture. The control side contributes most where the window is read as a sequence and least where the model reads one feature vector per scored minute, carrying each channel's last observation forward, so it sees a level and has little shape to drift with. The ordering is not clean: the tabular families' control means fall in MOVER (EPIC) as well and only flatten in the two external cohorts. Representation and family are not crossed in this design, so this is a reading the data supports rather than a hypothesis it tests. It does say where

to look: **the control-side contribution is a thing to measure in models given freer access to the input series**, and reporting the two sides of a widening gap separately is what makes it visible.

Three things this figure is not. The control series is aligned on the sampled pseudo-onset of Section 2 rather than on an event, so its fall is behavior over the last hour of a comparable window and not an approach to anything. Figure A5 plots medians and this figure means, and the two are not interchangeable here. And the two figures use different alignments: Figure A5's axis is elapsed time from the case's first scored minute and covers the first ten minutes of the record, where the transformer has just settled above its own 0.894 threshold, whereas the bins here sit far later in the case, where its event mean is 0.383. **The two are not in conflict, and the length of the plateau is what reconciles them.** The median score holds at or above the transported threshold through elapsed minute 24, 21 and 17 in event cases and 23, 20 and 16 in controls, then falls to 0.03 to 0.31 by minute 30. Figure A5 covers the first ten of those minutes and stops inside the plateau; every bin in this figure falls after it in all but the shortest records.

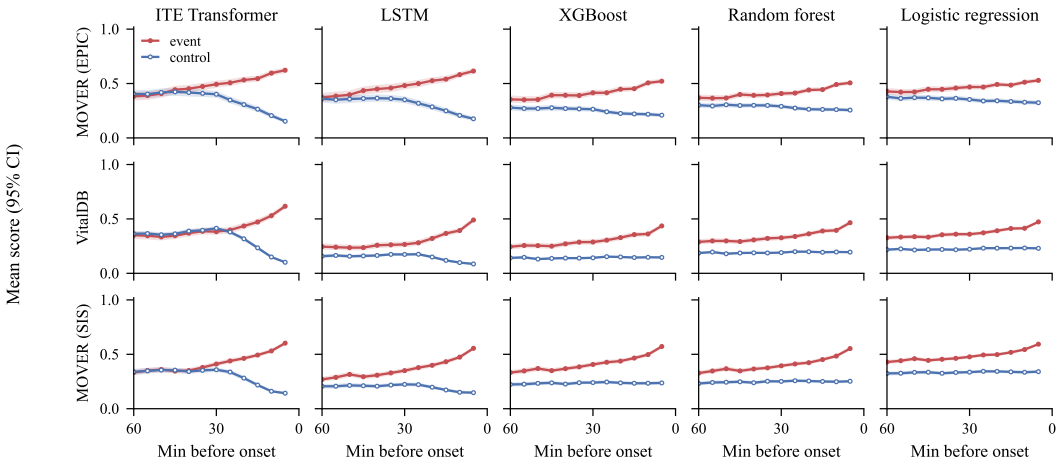


Figure A4: **Mean score in event and control cases against minutes before onset.** Rows are cohorts, columns are model families, the same grid as Figure A5, on the event-aligned axis instead of the elapsed one. Event cases solid and filled, control cases dashed and open; bands are 95% cluster bootstrap. Bins are the 13 centers at five-minute spacing from 60 minutes before onset; the bin at onset is empty by construction, because no minute is strictly before its own reference time, so the series ends at 5 minutes. Control cases are aligned on a sampled pseudo-onset.

A.14 Score against elapsed monitoring time

Section 4.1 reads the first ten monitored minutes of every case, and reports that only the ITE Transformer clears its own threshold there. The margins for the others are wide. Over those ten minutes the highest median score reached in either group is 0.69 against a threshold of 0.78 for logistic regression in MOVER (EPIC), 0.65 against a threshold of 0.91 for XGBoost, and 0.54 against a threshold of 0.79 for random forest; the LSTM clears its threshold in MOVER (EPIC) event cases and nowhere else. This is that view for all fifteen cells, on the elapsed axis rather than the event-aligned one of Figure A4.

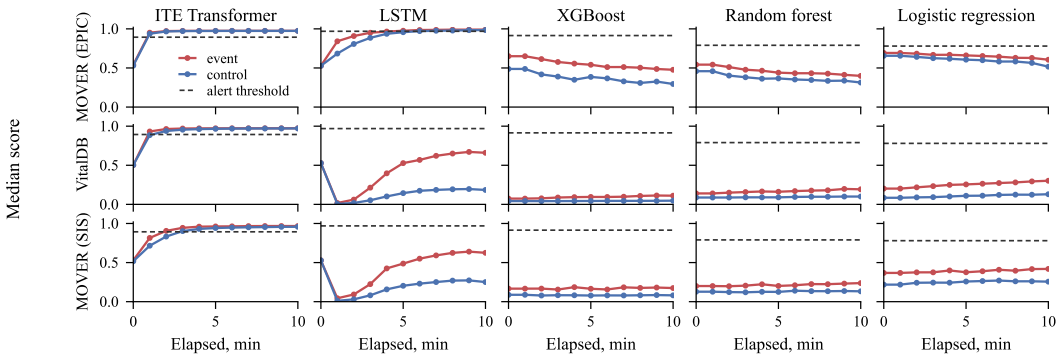


Figure A5: **Median score against elapsed monitoring time, with each family's own alert threshold.** Rows are cohorts, columns are model families. Elapsed time is measured from the case's first scored minute. The dashed rule in each panel is that family's transported raw-score threshold at the 80%-sensitivity target, transferred unchanged from the MOVER (EPIC) validation split. Only the ITE Transformer's early rise clears its own threshold, and it does so in event and control cases together. The elevation ends around minute 20, past the right edge of this axis. This axis is elapsed time; the event-aligned view is Figure A4.

A.15 Departure timing: searched for, not found

We asked whether the score departs from its own baseline before mean arterial pressure departs from its, within an episode, under a pre-specified rule: baseline is the median of the episode's first ten eligible minutes, the robust scale is 1.4826 $\times$ the median absolute deviation over the same minutes, and a departure is the first minute beginning a run of at least three consecutive minutes more than 1.5 scale units in the deteriorating direction.

Score departure does not precede MAP departure. For XGBoost the share of episodes in which the score departs first is significantly *below* one half in all three cohorts. For the transformer the score does not depart at all in most episodes, because its baseline sits at the top of its range: a median baseline of 0.94 to 0.97 leaves median headroom of 0.005 to 0.031, which is less than the bar its own rule sets in all three cohorts, so there is not enough room above baseline for a departure to register. Both facts are reported so the negative cannot be read as an unfair rule. Sensitivity analyses at a 2.0 scale threshold, a five-minute run length, and a search capped at the 60-minute window all leave the direction unchanged.

The question was pre-specified and the answer constrains Section 4: whatever produces the reported lead, it is not the score moving before the monitored variable does.

A.16 Lead time by band, the transformer's three cohorts

Table A9 | Median alert lead time inside quantile-derived bands of observed window, ITE Transformer. ITE Transformer, three held-out cohorts, at the transported 80%-sensitivity operating point. Quintile edges are cut within each cell on detected cases; where the observed window is censored at the 60-minute alert window the upper quintiles collapse and the realized band count is reported rather than assumed. `Observed window, min` is the median observed pre-onset window of that band's cases rather than an edge. Intervals are 95% cluster bootstrap over subjects, or over cases in MOVER (SIS). All fifteen cells are Table A1; the detection rate these bands condition on is Table A5. Section 3.1 reports the contrast this table carries.

Cohort	Band	n	Observed window, min	Median lead, min (95% CI)	Lead / window (95% CI)
MOVER (EPIC) test	1	139	10	9 [8, 9]	0.88 [0.85, 0.89]
MOVER (EPIC) test	2	111	19	18 [17, 18]	0.94 [0.94, 0.95]
MOVER (EPIC) test	3	121	33	31 [29, 32]	0.97 [0.96, 0.97]
MOVER (EPIC) test	4	240	60	46 [44, 50]	0.98 [0.96, 0.98]
VitalDB	1	143	7	5 [5, 6]	0.83 [0.83, 0.83]
VitalDB	2	140	12	10 [10, 11]	0.91 [0.90, 0.91]
VitalDB	3	132	20	19 [18, 20]	0.94 [0.94, 0.95]
VitalDB	4	134	40	38 [37, 40]	0.97 [0.97, 0.97]
VitalDB	5	135	60	59 [57, 60]	0.98 [0.98, 1.00]
MOVER (SIS)	1	199	8	6 [5, 6]	0.80 [0.80, 0.82]
MOVER (SIS)	2	179	17	14 [13, 15]	0.86 [0.84, 0.91]
MOVER (SIS)	3	186	29	27 [26, 28]	0.94 [0.93, 0.96]
MOVER (SIS)	4	362	60	46 [43, 49]	0.95 [0.93, 0.97]

A.17 Cohort composition and the control side

Table A10 | Cohort composition, PPV against prevalence, and the control side. `PPV − prevalence` is the lift over the base rate of the cohort the PPV is measured in, because raw PPV is not comparable across cohorts whose prevalence runs from 0.359 to 0.560. `Alerted by elapsed min 1` is one minus the Kaplan-Meier survival of time to first alert at the first elapsed monitoring minute, censored at the end of each case's own window; it is the censoring-aware form of Section 4.1's settling. Intervals are 95% cluster bootstrap.

Model	Cohort	Event cases	Control cases	Prevalence	PPV − prevalence	Min. above thr., controls (95% CI)	Alerted by elapsed min 1 (95% CI)
ITE Transformer	MOVER (EPIC) test	786	618	0.560	+0.065	0.150 [0.050, 0.233]	0.532 [0.497, 0.565]
ITE Transformer	VitalDB	1007	1797	0.359	+0.021	0.183 [0.163, 0.233]	0.460 [0.428, 0.491]
ITE Transformer	MOVER (SIS)	1409	1232	0.534	+0.051	0.025 [0.017, 0.047]	0.308 [0.285, 0.332]
LSTM	MOVER (EPIC) test	786	618	0.560	+0.101	0.017 [0.000, 0.027]	0.228 [0.198, 0.257]
LSTM	VitalDB	1007	1797	0.359	+0.236	0.000 [0.000, 0.000]	0.087 [0.071, 0.105]
LSTM	MOVER (SIS)	1409	1232	0.534	+0.151	0.000 [0.000, 0.000]	0.138 [0.120, 0.157]
XGBoost	MOVER (EPIC) test	786	618	0.560	+0.165	0.000 [0.000, 0.000]	0.188 [0.161, 0.215]
XGBoost	VitalDB	1007	1797	0.359	+0.253	0.000 [0.000, 0.000]	0.062 [0.047, 0.077]
XGBoost	MOVER (SIS)	1409	1232	0.534	+0.180	0.000 [0.000, 0.000]	0.119 [0.101, 0.136]
Random forest	MOVER (EPIC) test	786	618	0.560	+0.143	0.000 [0.000, 0.000]	0.183 [0.157, 0.212]
Random forest	VitalDB	1007	1797	0.359	+0.221	0.000 [0.000, 0.000]	0.101 [0.082, 0.121]
Random forest	MOVER (SIS)	1409	1232	0.534	+0.154	0.000 [0.000, 0.000]	0.166 [0.147, 0.186]
Logistic regression	MOVER (EPIC) test	786	618	0.560	+0.128	0.000 [0.000, 0.017]	0.226 [0.198, 0.255]
Logistic regression	VitalDB	1007	1797	0.359	+0.237	0.000 [0.000, 0.000]	0.036 [0.025, 0.048]
Logistic regression	MOVER (SIS)	1409	1232	0.534	+0.152	0.000 [0.000, 0.000]	0.121 [0.104, 0.138]

A.18 Multiplicity

The two nulls of Section 3.3 form one Benjamini–Hochberg family of 75 formal tests, adjusted within that family, of which 57 are significant at $q < 0.05$. The stratification of Section 3.1 produces intervals rather than tests and does not enter the family.

A.19 Seeds and reproducibility

Cluster bootstraps use 2,000 replicates and permutation nulls 1,000, with the clustering unit the subject for two cohorts and the case for MOVER (SIS), whose released subject key is a copy of the case identifier. Model training and the reference-time construction are fixed at a single seed, reported in Appendix A.1.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state the four contributions and their scope. They distinguish the observed lead–window association from what the upward-biased shuffle can establish, treat low directedness as descriptive rather than causal, and state that the paper measures neither clinician comprehension nor physiology.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

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